
Complements of Operational Research

Exercise sheet 7 — Inventory management

The exercises marked **homework** are to be solved at home and sent by email to the lecturer, by the deadline given in class; the solution is shared afterwards. The others are from previous tests and exams. ABC classification: A up to 80% of the cumulative value, B up to 95%, C the rest.

Exercise 1 — ABC classification

Homework (2nd test, 27/06/2019)

Carry out an ABC classification of the products in the following table.

Product	A	B	C	D	E	F	G	H	I	J
Unit value	1.40	0.60	4.20	5	3	0.16	2.25	20	2.80	5
Quantity	400	5700	1300	600	2600	24000	3600	289	600	58

Exercise 2 — Inventory with shortages (annual)

Homework

Time unit: year. Demand: 600 units/year; holding rate 20%; unit purchase cost 300 m.u.; ordering cost 50 m.u.; service level (ρ) 95%.

- Economic order quantity.
- Active (maximum) stock.
- Stock cycle period.
- Sub-period of positive stock.
- Sub-period of negative stock.
- Annual purchase cost.
- Annual ordering cost.
- Annual holding cost.
- Annual shortage cost.
- Total inventory cost per year.

Exercise 3 — Item AXX with shortages

Homework (exam, 10/07/2018)

Consider item AXX, with an average monthly demand (25 days) of 150 units and negligible variability. The capital needed to carry the stock for one month, with shortages, could alternatively be invested at a return of 10%. With a unit purchase cost of 10 m.u., an ordering cost of 50 m.u., a unit shortage cost of 2 m.u. and a lead time of 5 days, compute the optimal inventory management policy:

- Service level.
- Economic order quantity.
- Maximum (active) stock.
- Period (cycle).
- Time with active stock.
- Time out of stock.
- Reorder point.
- Monthly cost.
- Number of orders per month.
- Optimal policy to follow.

Exercise 4 — Coffee beans for a chain of coffee shops

Homework (new)

A chain of coffee shops uses 24 000 kg of coffee beans per year, at a steady rate. The roaster sells at 9 €/kg; each order costs 120 € (transport and receiving); the holding rate is 20% per year. The roaster offers a 3% discount on orders of 2000 kg or more.

- Find the economic order quantity without discount, the number of orders per year and the annual cost.
- Does it pay to take the discount? What is the optimal policy and its annual cost?
- Actual demand turned out to be 20% higher than forecast. Is the policy of part b) still optimal? And if the chain had kept the EOQ of part a), without discount, how much more would it pay per year compared with the optimal policy for the new demand? Comment.

Exercise 5 — Craft beer: producing in batches

Homework (new)

A craft brewery sells 48 000 L per year of its main beer, at a steady rate. The production and bottling line, if it works only on this beer, produces 80 000 L per year. Each batch requires cleaning and a recipe change, which cost 300 € and take 1 week. The production cost is 2.5 €/L and the holding rate is 20% per year. Consider a year of 50 working weeks.

- Find the optimal batch, the maximum stock, the production time of each batch, the cycle and the number of batches per year.
- What is the production launch point, that is, the stock level at which the set-up of the batch should start?
- Compute the annual cost, including production.
- A partner offers to supply the same beer at the same price, delivering each order all at once, at a cost of 300 € per order. What would the optimal batch and the annual management cost be? Explain the difference.
- For what fraction of the year is the line busy with this beer? What use can the remaining time have?

Exercise 6 — Item AAA with shortages

2nd test and exam, 27/06/2019

Consider item AAA, with an average monthly demand (25 days) of 180 units and negligible variability. The capital needed to carry the stock for one month, with shortages, could alternatively be invested at a return of 10%. With a unit purchase cost of 10 m.u., an ordering cost of 100 m.u., a unit shortage cost of 4 m.u. and a lead time of 5 days, compute the optimal inventory management policy:

- Service level.
- Economic order quantity.
- Maximum (active) stock.
- Period (cycle).
- Time with active stock.
- Time out of stock.
- Reorder point.
- Monthly cost.
- Number of orders per month.
- Optimal policy to follow.

Exercise 7 — Quantity discounts (I)

exam, 05/07/2021

A company buys abroad a certain type of component, of which it uses, on average, 18 000 units per year. A supplier offers to sell each component at 140 m.u., with the discounts shown. Processing an order costs 100 m.u. and the holding rate is 25% per year. Assume a year has 52 weeks.

Quantity	Discount
(0, 1999)	0%
(2000, 2999)	2%
(3000, 4999)	3%
(5000, 9999)	4%
$\geq 10\,000$	5%

- a) Economic order quantity.
- b) Period (in weeks).
- c) Annual cost.
- d) Number of orders per year.

Exercise 8 — Quantity discounts (II)

2nd test, 11/06/2024

A company buys abroad a certain type of component, of which it uses, on average, 25 000 units per year. A supplier offers to sell each component at 100 m.u., with the discounts shown. Processing an order costs 110 m.u. and the holding rate is 25% per year. Assume a year has 52 weeks.

Quantity	Discount
(0, 1999)	0%
(2000, 2999)	2%
(3000, 4999)	3%
(5000, 9999)	6%
$\geq 10\,000$	8%

- a) Economic order quantity.
- b) Period (in weeks).
- c) Annual cost.
- d) Number of orders per year.